

CME396 – PROCESS PLANNING AND COST ESTIMATION

CIAT – II (October 2026) • SET B • Answers / Study Material

Units 3, 4 and 5 • III Year / V Semester • Mechanical Engineering

PART A – 2 MARK ANSWERS

1. Why is costing and estimation important in manufacturing?

- To fix the selling price and submit quotations / tenders.
- To decide make-or-buy and choose the most economical process.
- To prepare budgets, control cost and find the actual profit.

2. List out the various methods of allocation of overhead expenses.

(i) Percentage of direct material cost, (ii) Percentage of direct labour cost, (iii) Percentage of prime cost, (iv) Man-hour rate, (v) Machine-hour rate, (vi) Combination of man-hour and machine-hour rates, (vii) Unit rate method.

3. Define man hour rate and machine hour rate.

$$\text{Man-hour rate} = \frac{\text{Total factory overheads}}{\text{Total man-hours worked}}$$

$$\text{Machine-hour rate} = \frac{\text{Total cost of running the machine}}{\text{Total machine hours}}$$

Man-hour rate is used for manual work; machine-hour rate is used where work is mainly done on machines.

4. Brief the term administrative expenses.

Administrative expenses are the costs of running the office (not the shop floor): salaries of manager and office staff, office rent, stationery, telephone, postage, legal and audit fees. They are added to factory cost to get production (office) cost.

5. List the different losses incurred in a forging shop.

(i) Shear (cutting) loss, (ii) Scale loss, (iii) Flash loss, (iv) Tonghold loss, (v) Sprue loss, (vi) Burr (punching) loss.

6. Distinguish between welding and soldering.

Welding	Soldering
Base metals are melted and fused.	Base metals are not melted; only the filler (solder) melts.
High temperature (above 1500 °C for steel).	Low temperature (below 450 °C).
Strong, permanent joint.	Weak joint; used for electrical and sealing work.

7. What is tonghold loss and scale loss in forging?

Tonghold loss: the extra length of bar held by the tongs during forging, which is cut off afterwards (about 1 to 1.5 × bar diameter).

Scale loss: the metal lost as iron oxide scale formed on the surface during heating; taken as about 6% of net weight for the first heating.

8. What are the factors to be considered when estimating jobs in a forging shop?

- Net weight of forging and the losses (scale, flash, tonghold, shear, sprue) → gross weight and stock size.
- Cost of material, heating (fuel / power), forging labour and die cost.
- Number of heats, type of forging (hand / drop / press) and overheads.

9. Why is machining time calculation important in manufacturing processes?

- To estimate labour cost and machine-hour cost of a job.
- To fix delivery dates, plan production and load machines.
- To fix incentive (bonus) rates and check worker efficiency.

10. Write down the formulae for calculating machining time for milling operations.

$$\text{Table feed } f_m = f_t \times Z \times N \text{ mm/min} \quad T = \frac{L + A + O}{f_m} \text{ min}$$

f_t = feed per tooth, Z = number of teeth, N = cutter rpm, L = length of job, O = over-travel. Approach for slab milling $A = \sqrt{d(D - d)}$; for face milling (cutter centred) $A = D/2$.

PART B – 13 MARK ANSWERS

11 (a) Explain the elements of cost and the ladder of cost with a neat sketch.

Elements of cost

1. **Material** – *Direct material* becomes part of the product (steel bar, castings, bought-out parts). *Indirect material* helps production but cannot be traced to a job (oil, cotton waste, small tools).
2. **Labour** – *Direct labour* works directly on the product (turner, welder, fitter). *Indirect labour* includes supervisors, storekeepers, inspectors.
3. **Expenses** – *Direct expenses* are chargeable to one job (special jigs, patterns, royalty, hire of special machine). *Indirect expenses* (overheads) include rent, power, lighting, depreciation, insurance.

Overheads = Indirect material + Indirect labour + Indirect expenses; classified as factory, administrative and selling & distribution overheads.

Ladder of cost

Prime cost = Direct material + Direct labour + Direct expenses
Factory (works) cost = Prime cost + Factory overheads
Production (office) cost = Factory cost + Administrative overheads
Total cost = Production cost + Selling & distribution overheads
Selling price = Total cost + Profit

Prime cost	Direct material	Direct labour	Direct expenses
Factory cost	Prime cost		Factory overhead
Production cost	Factory cost		Admin. overhead
Total cost	Production cost		Selling & distribution
Selling price	Total cost		Profit

Fig. Ladder of cost

11 (b) Explain the steps involved in estimating labour and material costs in manufacturing.

A. Material cost

1. Divide the component into simple shapes (cylinders, prisms, cones) from the drawing.
2. Find the volume of each and add them (subtract holes).
3. Weight = Volume $\times$ Density (steel 7.86 g/cm³, cast iron 7.2 g/cm³, aluminium 2.7 g/cm³).
4. Add allowances – machining allowance, scrap, cutting (kerf) loss, scale loss.
5. Material cost = Gross weight $\times$ Rate per kg + bought-out parts – scrap value.

B. Labour cost

1. List the operations from the operation sheet.
2. Find set-up time, operation (machining) time and handling time for each operation.
3. Add allowances (10–20%) for fatigue, personal needs and delays.

$$\text{Time per piece} = \frac{\text{Set-up time}}{\text{Batch size}} + \text{Operation time} + \text{Handling time} + \text{Allowances}$$

$$\text{Labour cost} = \text{Total time} \times \text{Wage rate per hour}$$

Example

Shaft: 12 min turning + 3 min handling + 15% allowance = $15 \times 1.15 = 17.25$ min. At Rs. 120 per hour, labour cost = $(17.25/60) \times 120 = \text{Rs. 34.50}$. Material: $\varnothing 40 \times 200$ mm steel bar $\rightarrow$ volume = $251.3 \text{ cm}^3 \rightarrow$ weight = 1.975 kg $\rightarrow$ at Rs. 80/kg = **Rs. 158**.

12 (a) Discuss the various methods of costing, such as job costing, process costing and activity-based costing, highlighting their differences and suitability.

1. **Job costing** – cost found separately for each job / order; each job has a cost sheet. Suited to job shops, repair shops, special machines.
2. **Batch costing** – a batch of identical items is one job; cost per item = batch cost $\div$ number of items. Bolts, gears, medicines.
3. **Process costing** – product passes through processes in sequence; cost of each process is passed to the next. Cement, sugar, steel, chemicals.
4. **Unit costing** – one standard product in bulk (bricks, cement bags, mining).
5. **Operating costing** – services: transport (per km), power (per kWh).
6. **Contract costing** – long works: buildings, bridges, ships.
7. **Activity-based costing (ABC)** – overheads are traced to activities (set-ups, inspection, material handling) through *cost drivers* and charged to products according to their use of each activity.

Point	Job costing	Process costing	Activity-based costing
Type of production	Made to order, different each time	Continuous, identical products	Many products, high overheads
Cost unit	Job / order	Process / period	Activity
Overhead charging	Broad (blanket) rate	Per process	Per activity cost driver
Accuracy	Moderate	Average cost only	High
Example	Machine repair, tool room	Cement plant, refinery	Auto-parts plant with many models

Conclusion: job costing suits job-order shops, process costing suits continuous production and ABC gives the most accurate costs in multi-product factories.

12 (b) Elaborate the concept of appreciation and depreciation. Explain the methods of calculating depreciation with an example.

Depreciation

Permanent decrease in the value of an asset with time. **Causes:** wear and tear, age and decay, obsolescence, accidents, fall in demand.

Appreciation

Increase in the value of an asset with time, e.g. land and buildings in growing cities. A plot bought for Rs. 10 lakh may be worth Rs. 15 lakh after 5 years. Appreciation is not charged to product cost.

Methods – example: machine C = Rs. 1,00,000, S = Rs. 10,000, n = 5 years

1. Straight line method

$$D = \frac{C - S}{n} = \frac{1,00,000 - 10,000}{5} = \text{Rs. } 18,000 \text{ per year}$$

2. Reducing balance method

$$r = 1 - \left(\frac{S}{C}\right)^{1/n} = 1 - (0.1)^{0.2} = 1 - 0.631 = 0.369 = 36.9\%; \text{ 1st year } D = \text{Rs. } 36,900$$

3. Sinking fund method (i = 10%)

$$A = (C - S) \times \frac{i}{(1 + i)^n - 1} = 90,000 \times \frac{0.1}{(1.1)^5 - 1} = \text{Rs. } 14,742 \text{ per year}$$

4. Sum-of-years' digits (sum = 1 + 2 + 3 + 4 + 5 = 15)

$$D_k = (C - S) \times \frac{n - k + 1}{15}; \text{ 1st year } = 90,000 \times 5/15 = \text{Rs. } 30,000; \text{ 2nd year } = \text{Rs. } 24,000$$

5. Machine-hour basis (2,000 h/year → 10,000 h)

$$D \text{ per hour} = \frac{C - S}{\text{Total hours}} = \frac{90,000}{10,000} = \text{Rs. } 9 \text{ per hour}$$

13 (a) Explain the steps involved in estimating a job in a forging shop.

Steps

1. **Net weight** – volume of forging from the drawing × density.
2. **Add losses** – scale loss (≈ 6%), flash loss, tonghold loss, shear loss, sprue loss, burr loss.
3. **Gross weight** = Net weight + Losses.
4. **Stock size** – choose bar diameter; length of stock = Gross volume ÷ area of bar.
5. **Material cost** = Gross weight × Rate per kg.
6. **Labour cost** – forging time (heating + forging + trimming) × labour rate.
7. **Die / tool cost** per piece (drop forging) and heating (fuel) cost.
8. **Add overheads** → total forging cost.

Example

Forge a collar Ø60 × 20 mm on a shaft Ø30 × 120 mm from Ø40 mm bar. Density 7.86 g/cm³; scale loss 6%, flash loss 5%, tonghold length = 1 × bar diameter; steel Rs. 80/kg.

Item	Working	Volume (cm ³)
Collar	$\pi/4 \times 6^2 \times 2$	56.55
Shaft	$\pi/4 \times 3^2 \times 12$	84.82
Net volume		141.37
Scale loss 6%	0.06×141.37	8.48
Flash loss 5%	0.05×141.37	7.07
Tonghold loss	$\pi/4 \times 4^2 \times 4$	50.27
Gross volume		207.19

Gross weight = $207.19 \times 7.86 = 1,628 \text{ g} = \mathbf{1.63 \text{ kg}}$; length of Ø40 stock = $207.19 \div (\pi/4 \times 4^2) = 16.5 \text{ cm} = \mathbf{165 \text{ mm}}$; material cost = $1.63 \times 80 = \mathbf{Rs. 130.3}$.

13 (b) Illustrate the procedure for doing real time estimation in a modernized semi-automated foundry shop with a case example.

Meaning

In a semi-automated foundry (mechanised moulding line, induction furnace, ERP and sensors), cost data such as metal weight poured, power used and cycle time are captured live, so the cost of each casting is estimated and updated in real time.

Procedure

1. **CAD model of casting** – exact net weight from 3D model.
2. **Gating and feeding design** (simulation software) – weight of gates and risers, yield %.
3. **Metal charge** = poured weight $\div$ (1 – melting loss); metal cost from current market rate; credit for returned gates and risers.
4. **Melting cost** – kWh per kg from induction-furnace energy meter $\times$ tariff.
5. **Moulding and core cost** – cycle time of moulding machine $\times$ machine-hour rate; sand and binder cost.
6. **Pattern / tooling cost** spread over the order quantity.
7. **Fettling, shot blasting, inspection**; rejection allowance from live quality data.
8. **Overheads and profit**; the ERP updates the estimate as actual data comes in.

Case example – cast iron pump body, net weight 12 kg, order 1,000 nos.

Item	Working	Rs. / casting
Poured metal	12 kg + 25% gates & risers = 15 kg	
Metal charge	$15 \div 0.95$ (5% melting loss) = 15.79 kg $\times$ Rs. 60	947.4
Less: return scrap credit	3 kg $\times$ Rs. 40	– 120.0
Melting power	$15.79 \text{ kg} \times 0.6 \text{ kWh/kg} \times \text{Rs. 8}$	75.8
Pattern	$\text{Rs. 50,000} \div 1,000$	50.0
Moulding and cores	machine cycle cost + sand	90.0
Fettling and inspection		40.0

Prime cost		1,083.2
Overheads	30%	325.0
Total cost per casting		1,408.2

14 (a) Discuss the specific challenges and considerations involved in estimating production costs for different types of jobs in forging, welding and foundry shops.

Shop	Challenges	Considerations
Forging	Losses (scale, flash, tonghold) vary with shape and number of heats; die cost is high and die life uncertain; fuel cost changes.	Net and gross weight, stock size, number of heats, die cost per piece, forging time, type of forging (hand / drop / press).
Welding	Weld size and number of passes vary; electrode losses (stub ends, spatter); operator factor depends on skill and position; distortion and rework.	Joint design and edge preparation, weld metal volume, deposition efficiency, arc time, power / gas cost, fixtures.
Foundry	Casting defects and rejections; yield depends on gating design; pattern cost depends on quantity; melting losses; metal price fluctuation.	Casting weight, gates and risers, melting loss, pattern and core cost, moulding method, fettling, inspection.

Common challenges

- Non-standard, one-off jobs with little past data.
- Fluctuating material and energy prices.
- Variation in worker skill and machine condition.
- Allocation of high overheads (furnaces, presses) to individual jobs.

Remedies

Use standard data and past job records, CAD for exact volumes, simulation (casting / forging software), ERP for live prices, time study, and a fixed contingency for rejections.

14 (b) Welding cost: two MS plates 1 m long, 10 mm thick, single V butt (60°, root gap 2 mm). Density 7.8 g/cm³; deposition efficiency 70%; electrode Rs. 120/kg; speed 0.15 m/min; operator factor 50%; labour Rs. 200/h; overheads 100% of labour; 25 V, 200 A, $\eta = 60\%$; power Rs. 8/kWh.

1. Weld metal

$$A = g t + t^2 \tan(\theta/2) = 2 \times 10 + 10^2 \times \tan 30^\circ = 20 + 57.74 = 77.74 \text{ mm}^2$$

$$\text{Volume} = 77.74 \times 1000 = 77,740 \text{ mm}^3 = 77.74 \text{ cm}^3 \quad \text{Weight} = 77.74 \times 7.8 = 606.3 \text{ g} = 0.606 \text{ kg}$$

2. Electrode cost

$$\text{Electrode required} = \frac{0.606}{0.70} = 0.866 \text{ kg} \quad \text{Cost} = 0.866 \times 120 = \text{Rs. 103.94}$$

3. Labour and overheads

$$\text{Arc time} = \frac{1}{0.15} = 6.67 \text{ min} \quad \text{Total time} = \frac{6.67}{0.5} = 13.33 \text{ min} = 0.222 \text{ h}$$

$$\text{Labour} = 0.222 \times 200 = \text{Rs. 44.44} \quad \text{Overheads (100\%)} = \text{Rs. 44.44}$$

4. Power cost

$$\text{Energy} = \frac{V \times I}{1000 \times \eta} \times \text{arc time} = \frac{25 \times 200}{1000 \times 0.6} \times \frac{6.67}{60} = 0.926 \text{ kWh} \quad \text{Cost} = 0.926 \times 8 = \text{Rs. 7.41}$$

Cost element	Rs.
Electrode	103.94
Labour	44.44
Overheads	44.44
Power	7.41
Total cost of welding	200.23

Total welding cost $\approx$ Rs. 200 per metre of joint

15 (a) Illustrate the steps in finding the machining time for a sample component: (i) Drilling and tapping (6) (ii) Boring and taper turning (7)

(i) Drilling and tapping

Drilling – $\varnothing 20$ mm hole, 40 mm deep, $V = 25$ m/min, $f = 0.2$ mm/rev.

$$N = \frac{1000 \times 25}{\pi \times 20} = 397.9 \text{ rpm} \quad L = l + 0.3 d = 40 + 6 = 46 \text{ mm} \quad T = \frac{46}{0.2 \times 397.9} = \mathbf{0.578 \text{ min}}$$

Tapping – M12 $\times$ 1.75, 25 mm deep, $V = 6$ m/min. Feed = pitch; the tap must be reversed out, so time is doubled.

$$N = \frac{1000 \times 6}{\pi \times 12} = 159.2 \text{ rpm} \quad T = \frac{2 L}{p \times N} = \frac{2 \times 25}{1.75 \times 159.2} = \mathbf{0.180 \text{ min}}$$

(ii) Boring and taper turning

Boring – $\varnothing 40 \rightarrow \varnothing 50$ mm over 60 mm, depth 1 mm/pass, $V = 30$ m/min, $f = 0.2$ mm/rev.

$$\text{Cuts} = \frac{50 - 40}{2 \times 1} = 5 \quad N = \frac{1000 \times 30}{\pi \times 50} = 191 \text{ rpm} \quad T = \frac{60 \times 5}{0.2 \times 191} = \mathbf{7.85 \text{ min}}$$

Taper turning – $\varnothing 50 \rightarrow \varnothing 40$ over 80 mm axial length, depth 1 mm/pass, $V = 30$ m/min, $f = 0.25$ mm/rev; N at mean diameter 45 mm.

$$\text{Cuts} = \frac{50 - 40}{2 \times 1} = 5 \quad \text{Taper length} = \sqrt{(80^2 + 5^2)} = 80.16 \text{ mm} \quad N = \frac{1000 \times 30}{\pi \times 45} = 212.2 \text{ rpm}$$

$$T = \frac{80.16 \times 5}{0.25 \times 212.2} = \mathbf{7.55 \text{ min}}$$

15 (b) Explain the steps in calculating the machining time for shaping and planing operations with an illustrative numerical case example.

Steps

1. Length of stroke L = length of job + approach + over-travel (clearances).
2. Width to be covered B = width of job + side clearances.

3. Number of strokes = $B \div \text{feed per stroke}$.

4. Ratio $K = \text{return time} \div \text{cutting time}$ (or $V_{\text{cutting}} \div V_{\text{return}}$).

5. Time per double stroke = $\frac{L(1+K)}{1000 V}$ (or $L/V_c + L/V_r$).

6. Total time = Number of strokes $\times$ time per stroke $\times$ number of cuts.

In a **shaper** the tool reciprocates and the job is fed; in a **planer** the table (job) reciprocates and the tool is fed. The calculation is the same.

Example 1 – shaper

Plate 300×200 mm; stroke along 300 mm with 25 mm clearance each end; feed 0.5 mm/stroke; $V = 12$ m/min; return : cutting = 3 : 5 ($K = 0.6$).

$L = 350$ mm; strokes = $200 \div 0.5 = 400$; time/stroke = $\frac{350 \times 1.6}{1000 \times 12} = 0.0467$ min; $T = 400 \times 0.0467 =$
18.67 min

Example 2 – planer

Job 1500 mm long, 500 mm wide; over-travel 100 mm each end; cutting speed 20 m/min, return speed 40 m/min; feed 2 mm/stroke.

$L = 1.7$ m; strokes = 250; time/stroke = $1.7 (1/20 + 1/40) = 0.1275$ min; $T = 250 \times 0.1275 =$ **31.88 min**

PART C – 15 MARK ANSWERS

16 (a) Raw material $\text{Ø}45 \times 95$ mm. Face both ends to 90 mm; turn $\text{Ø}40$ full length; turn $\text{Ø}30$ over 70 mm and $\text{Ø}25$ over 45 mm from the same end; $45^\circ \times 3.5$ mm chamfer and thread (pitch 3 mm) over 35 mm on the $\text{Ø}25$ end; drill $\text{Ø}15 \times 20$ mm deep at the other end. $V_{\text{turn}} = 30$ m/min, $f = 0.35$ mm/rev, depth 1.25 mm; $V_{\text{drill}} = 30$ m/min, $f = 0.1$ mm/rev; chamfer feed 0.25 mm/rev; $V_{\text{thread}} = 10$ m/min.

Assumptions: approach and over-travel neglected; facing at N for $\text{Ø}45$; thread depth per cut 0.2 mm.

1. Facing (2.5 mm from each end → 2 cuts per end)

$$N = \frac{1000 \times 30}{\pi \times 45} = 212.2 \text{ rpm} \quad T = \frac{45/2}{0.35 \times 212.2} \times 2 \text{ cuts} \times 2 \text{ ends} = \mathbf{1.212 \text{ min}}$$

2. Turning $\text{Ø}45 \rightarrow \text{Ø}40$ over 90 mm (2 cuts)

$$T = \frac{90}{0.35 \times 212.2} \times 2 = \mathbf{2.424 \text{ min}}$$

3. Turning $\text{Ø}40 \rightarrow \text{Ø}30$ over 70 mm (4 cuts)

$$N = \frac{1000 \times 30}{\pi \times 40} = 238.7 \text{ rpm} \quad T = \frac{70}{0.35 \times 238.7} \times 4 = \mathbf{3.351 \text{ min}}$$

4. Turning $\text{Ø}30 \rightarrow \text{Ø}25$ over 45 mm (2 cuts)

$$N = \frac{1000 \times 30}{\pi \times 30} = 318.3 \text{ rpm} \quad T = \frac{45}{0.35 \times 318.3} \times 2 = \mathbf{0.808 \text{ min}}$$

5. Chamfering $45^\circ \times 3.5$ mm

$$N = \frac{1000 \times 30}{\pi \times 25} = 382.0 \text{ rpm} \quad T = \frac{3.5}{0.25 \times 382.0} = \mathbf{0.037 \text{ min}}$$

6. Drilling $\text{Ø}15 \times 20$ mm

$$N = \frac{1000 \times 30}{\pi \times 15} = 636.6 \text{ rpm} \quad T = \frac{20 + 0.3 \times 15}{0.1 \times 636.6} = \mathbf{0.385 \text{ min}}$$

7. Thread cutting, pitch 3 mm, length 35 mm

$$\text{Depth of thread} = 0.613 \times 3 = 1.84 \text{ mm} \rightarrow \text{cuts} = 1.84 \div 0.2 \approx 10 \quad N = \frac{1000 \times 10}{\pi \times 25} = 127.3 \text{ rpm}$$

$$T = \frac{35 \times 10}{3 \times 127.3} = \mathbf{0.916 \text{ min}}$$

Operation	Time (min)
Facing both ends	1.212
Turning Ø45 → Ø40	2.424
Turning Ø40 → Ø30	3.351
Turning Ø30 → Ø25	0.808
Chamfering	0.037
Drilling	0.385
Thread cutting	0.916
Total machining time	9.133

Total machining time ≈ 9.13 min per piece

16 (b) Discuss the significance of machining time calculation. How is machining time calculated for milling, shaping and grinding operations?

Significance

- Basis for labour cost and machine-hour cost in cost estimation.
- Fixing selling price and preparing quotations.
- Production planning, machine loading, delivery dates.
- Fixing incentive rates and comparing alternative processes and machines.

1. Milling

$$f_m = f_t \times Z \times N \text{ mm/min} \quad T = \frac{L + A + O}{f_m} \quad \text{Slab milling: } A = \sqrt{d(D - d)}$$

Example: $L = 150 \text{ mm}$, cutter $D = 80 \text{ mm}$, depth $d = 5 \text{ mm}$, $f_t = 0.1 \text{ mm}$, $Z = 10$, $N = 100 \text{ rpm}$, $O = 5 \text{ mm} \rightarrow f_m = 100 \text{ mm/min}$; $A = \sqrt{(5 \times 75)} = 19.4 \text{ mm}$; $T = (150 + 19.4 + 5) \div 100 = \mathbf{1.74 \text{ min}}$.

2. Shaping

$$T = \frac{B}{f} \times \frac{L(1 + K)}{1000 V}$$

Example: $L = 350 \text{ mm}$, $B = 200 \text{ mm}$, $f = 0.5 \text{ mm/stroke}$, $V = 12 \text{ m/min}$, $K = 0.6 \rightarrow T = 400 \times 0.0467 = \mathbf{18.67 \text{ min}}$.

3. Grinding (cylindrical, traverse)

$$\text{Traverse feed } f = \left(\frac{1}{2} \text{ to } \frac{3}{4}\right) \times \text{wheel width per rev} \quad \text{Passes} = \frac{\text{Allowance on radius}}{\text{Depth per pass}} \quad T = \frac{L}{f \times N_w} \times \text{Passes}$$

Example: $L = 200 \text{ mm}$, wheel width $25 \text{ mm} \rightarrow f = 12.5 \text{ mm/rev}$; work speed $N_w = 100 \text{ rpm}$; allowance 0.2 mm on diameter (0.1 mm on radius), depth $0.02 \text{ mm/pass} \rightarrow 5 \text{ passes}$; $T = 200 \div (12.5 \times 100) \times 5 = \mathbf{0.80 \text{ min}}$.

Conclusion

Accurate machining time is the foundation of correct costing and planning; each operation uses its own formula built on speed, feed, length of cut and number of passes.